

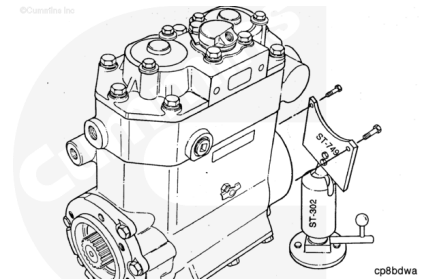
Trainings ▾

Disassemble

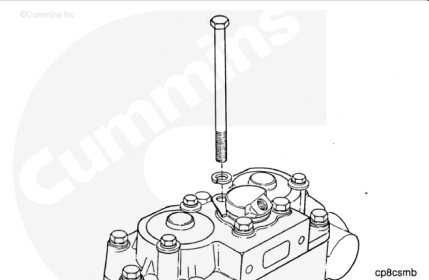
⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

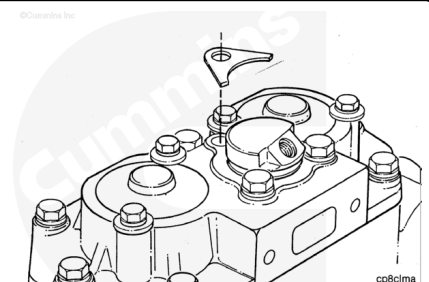
Use two 5/16 - 18 x 1 1/4 inch capscrews to mount the air compressor on the mounting plate, Part Number ST-749, or equivalent, which is used with the ball joint vise, Part Number ST-302, or equivalent.



Remove the center unloader body hexagon head capscrew and lock washer.



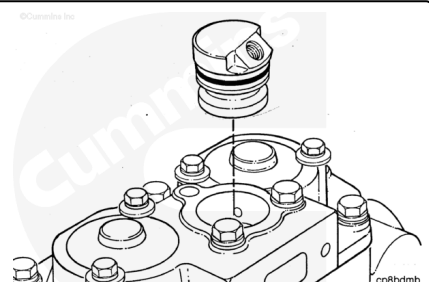
Remove the retaining clamp.



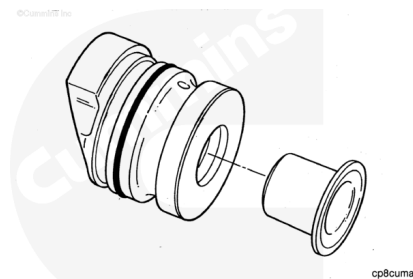
⚠ CAUTION ⚠

The unloader valve body is installed with spring tension. Carefully remove to avoid personal injury. Wear face and eye protection.

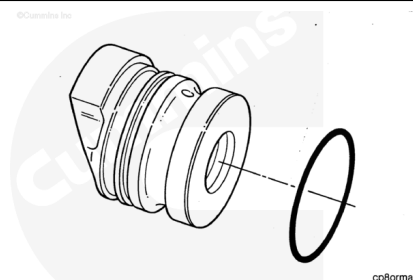
Remove the unloading valve body.



Remove the unloader valve cap, if installed.

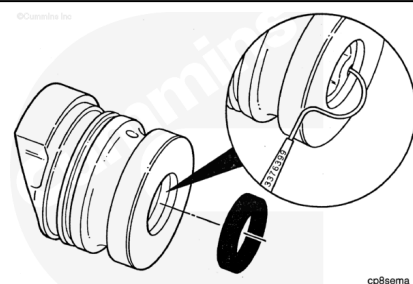


Remove and discard the o-ring seal.

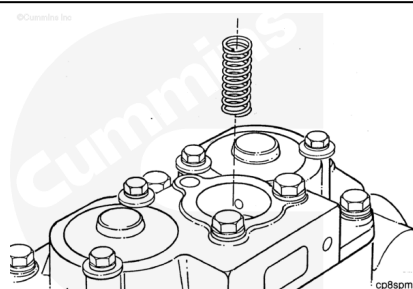


Use an o-ring pick, Part Number 3376399, or equivalent, to remove the rectangular ring seal, if installed.

Discard the seal.

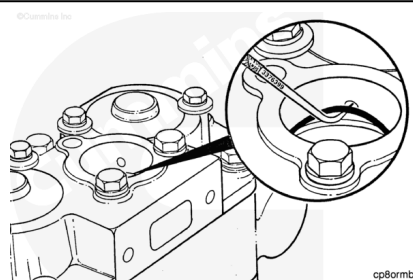


Remove the unloader valve cap spring.

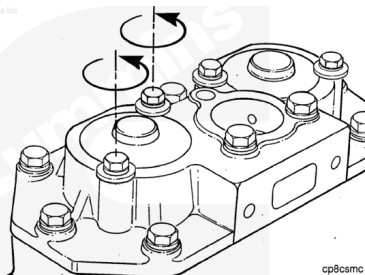


Use a o-ring pick, Part Number 3376399, or equivalent, to remove the o-ring seal.

Discard the seal.



Loosen the two unloader body hexagon head capscrews.

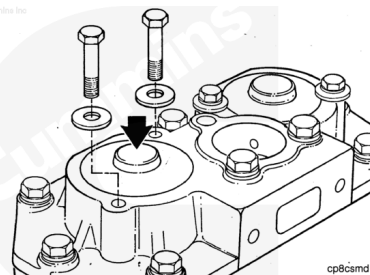


cp8csmc

⚠ WARNING ⚠

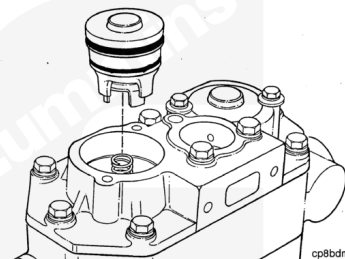
The unloader body is installed with spring tension, it must be removed carefully to avoid personal injury. Always wear protective eye wear.

Hold down on the unloader body and slowly remove the two unloader body capscrews and the two plain washers so as to prevent the unloader body from being thrown free and causing personal injury.



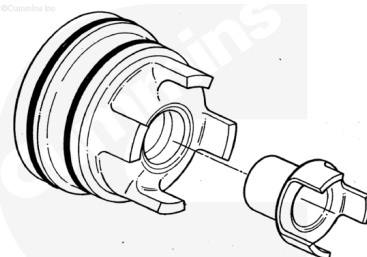
cp8csmc

Remove the unloader valve body.



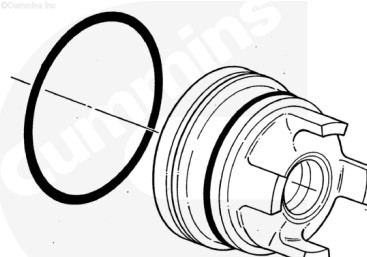
cp8bdme

Remove the unloader cap.



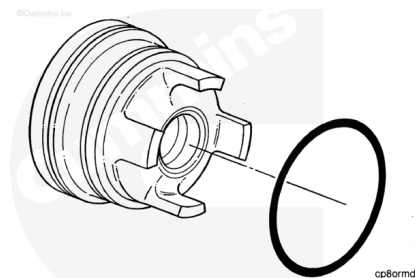
cp8cumb

Remove and discard the top o-ring seal.



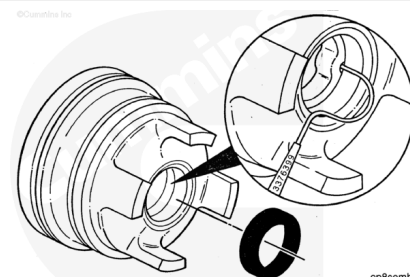
cp8ormc

Remove and discard the bottom o-ring seal.



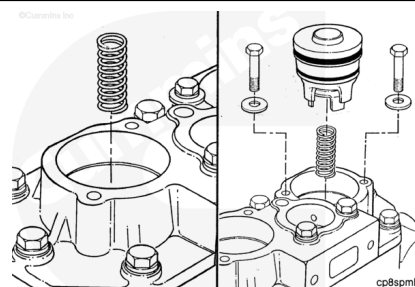
Use an o-ring pick, Part Number 3376399, or equivalent, to remove the rectangular ring seal.

Discard the seal.



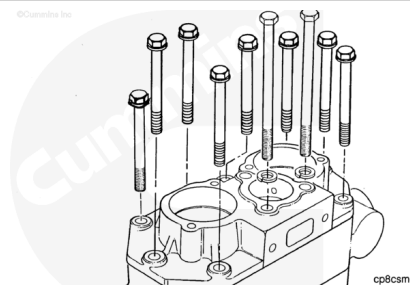
Remove the unloader cap spring.

Repeat the last eight steps to remove the other unloader body assembly.

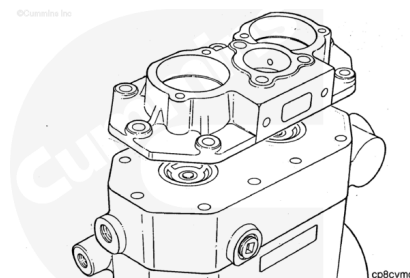


For the Holset® ST676 air compressor, remove the ten remaining hexagon head capscrews and two lock washers.

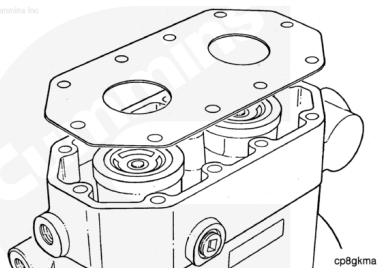
For the Holset® ST773 air compressor, remove the eight captive washer capscrews and the two hexagon head capscrews and lock washers.



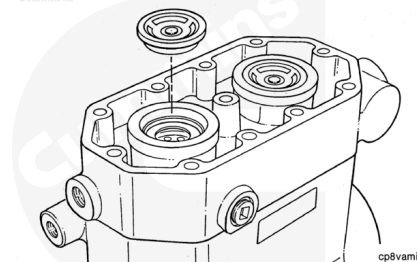
Remove the cover.



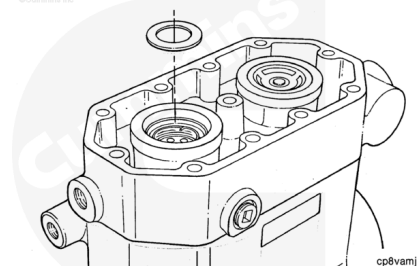
Remove and discard the cover gasket.



Remove the intake valve seat.

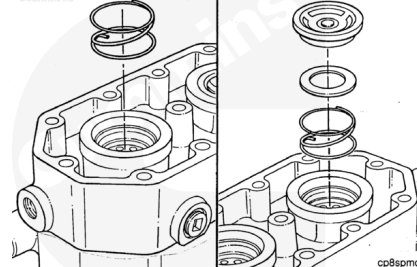


Remove the intake valve.

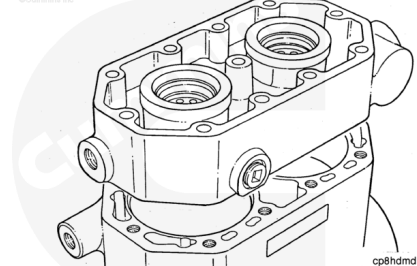


Remove the intake valve spring.

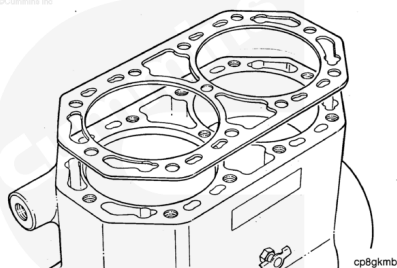
Repeat the last three steps to remove the other intake valve assembly.



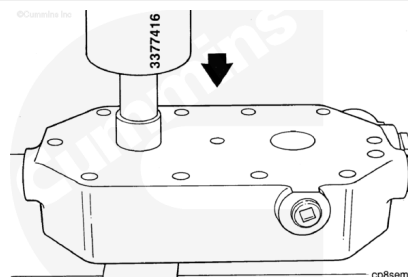
Remove the head.



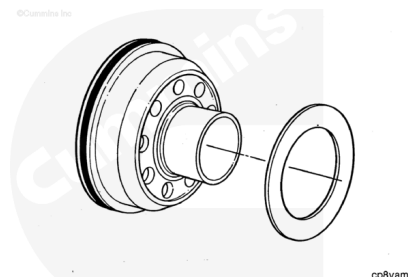
Remove and discard the head gasket.



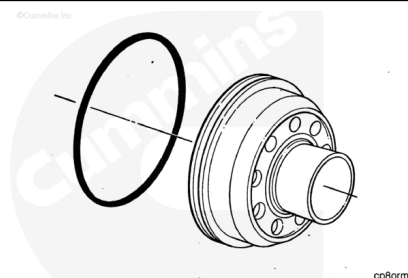
Use a press and an air compressor seat removal tool, Part Number 3377416, or equivalent, to remove the exhaust valve seat assembly from the head.



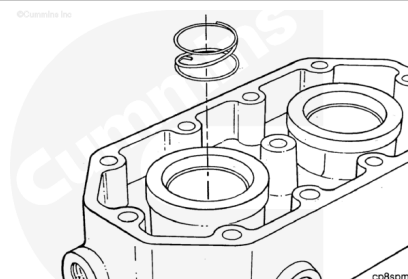
Remove the exhaust valve.



Remove and discard the o-ring seal.

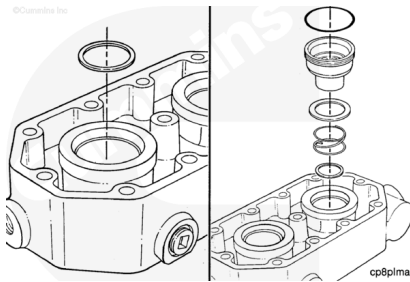


Remove the exhaust valve spring.



Remove the wear plate.

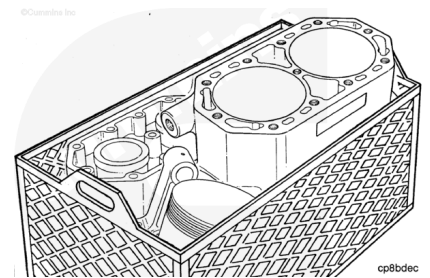
Repeat the last five steps to disassemble the other exhaust valve seat assembly.



Clean and Inspect for Reuse

⚠ WARNING ⚠

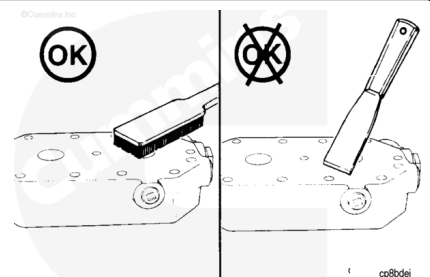
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing.



Soak the parts in a kerosene emulsion based cleaner designed to remove carbon. The cleaner **must** have a pH of 9.5 or less to avoid turning aluminum parts black. The cleaner manufacturer or supplier **must** be contacted about solution concentration, temperature and soak time.

⚠ CAUTION ⚠

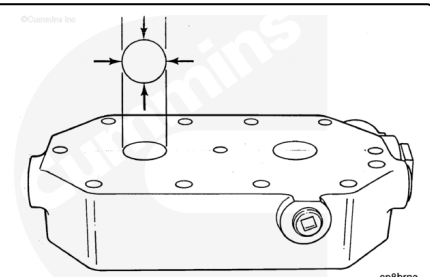
Do not use a scraper to remove carbon and scale, the sealing surfaces can be damaged.



The parts can be scrubbed with a stiff nonmetallic bristle brush.

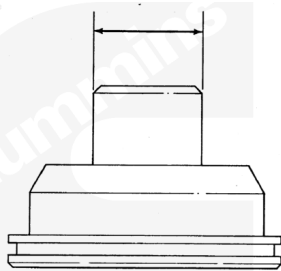
Measure the exhaust valve seat press fit diameter in the cylinder head.

Replace if worn larger than 20.688 mm [0.8145 in].



Measure the exhaust valve seat press fit diameter.

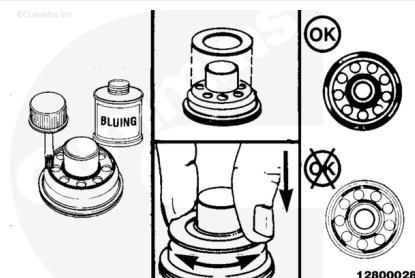
Replace if worn smaller than 20.714 mm [0.8155 in].



cp8sena

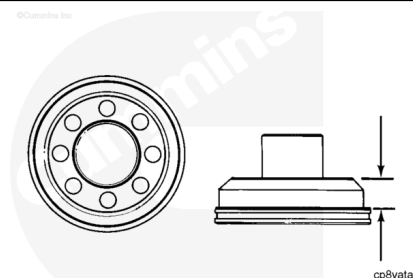
Apply "bluing" to the exhaust seating surface to check the seat.

If the seating surface is **not** 100 percent true discard and replace the valve seat.



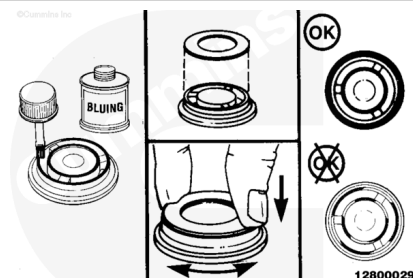
Measure the exhaust valve seat height.

If the height is less than 12.32 mm [0.485 in] replace the exhaust valve seat.



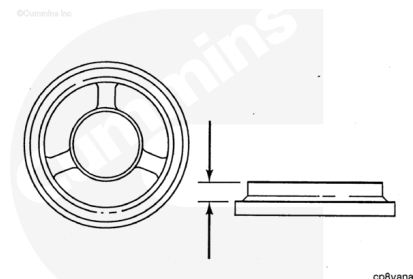
Apply "bluing" to the intake valve seating surface.

If the seating surface is **not** 100 percent true replace the intake valve seat.



Measure the intake valve seat height.

If the height is less than 6.86 mm [0.270 in] replace the intake valve seat.

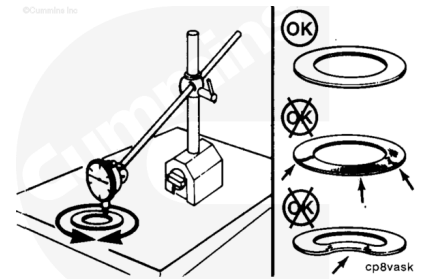


Note : Cummins Inc. recommends that new valves be installed during rebuild.

Inspect the exhaust and intake valves for cracks and damage.

Measure valves for flatness. The valves **must** be flat within 0.03 mm [0.001 in].

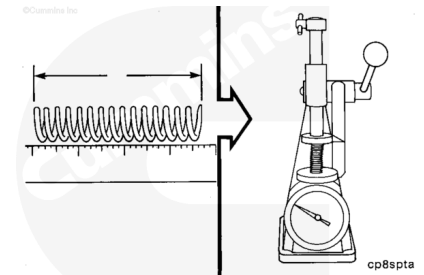
Replace the valves if cracked, damaged, or **not** flat.



Note : Cummins Inc. recommends that new springs be installed during rebuild.

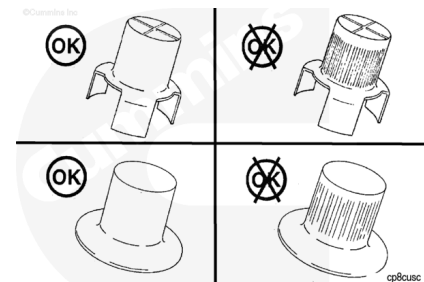
Use a valve spring tester, Part Number 3375182, or equivalent, to check the springs.

Replace if **not** within limits listed in the Compressor Spring Force Specifications table in Section V.



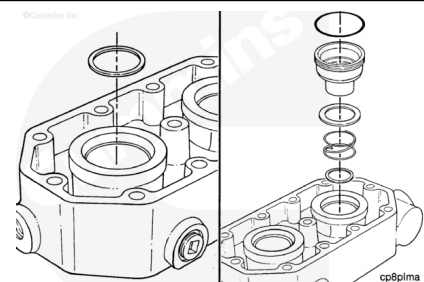
Inspect the upper part of the unloader cap where the rectangular ring seal seats for scoring.

Replace if scored.



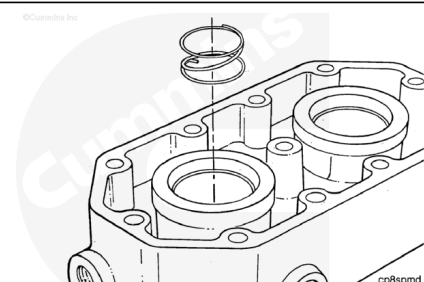
Assemble

Install the wear plate.

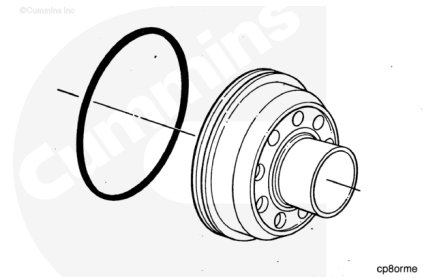


Note : Cummins Inc. recommends that new springs be installed during rebuild.

Install the exhaust valve spring with the tang end down.

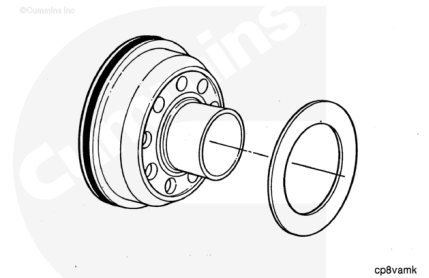


Install a new o-ring seal.

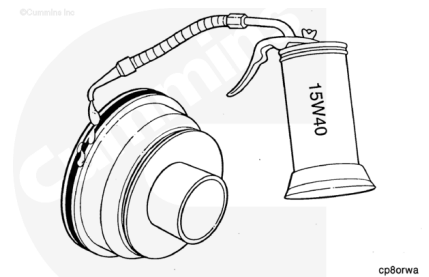


Note : Cummins Inc. recommends that new exhaust valves be installed during rebuild.

Install the exhaust valve.



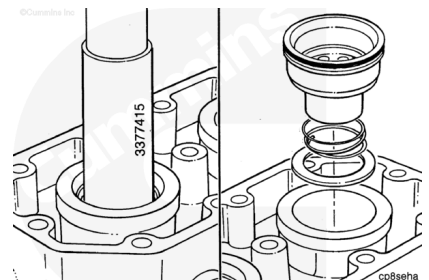
Use clean 15W-40 lubricating engine oil to lubricate the seal.



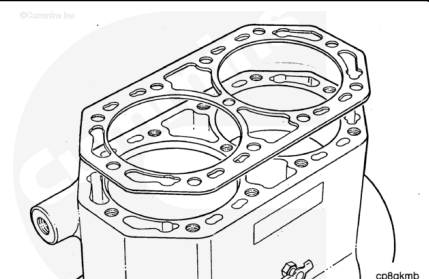
Note : Do not use excessive pressure on the exhaust valve seats, to do so can distort the valve.

Use a hand press and an air compressor seat installation tool, Part Number 3377415, or equivalent, to press the exhaust valve seat into the head.

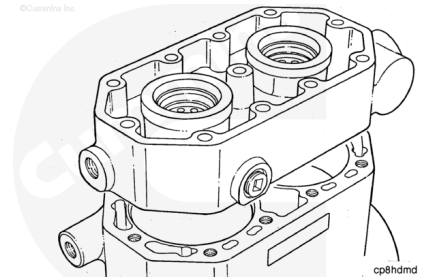
Repeat the last six steps to install the other exhaust valve assembly.



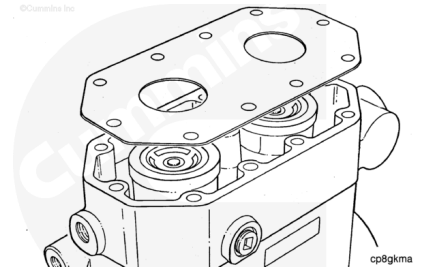
Install a new cylinder head gasket.



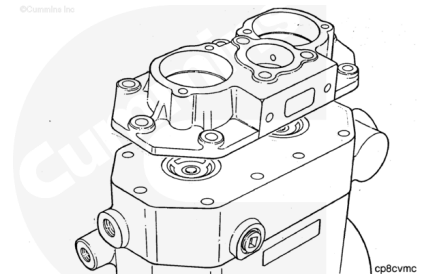
Install the head.



Install a new cover gasket.

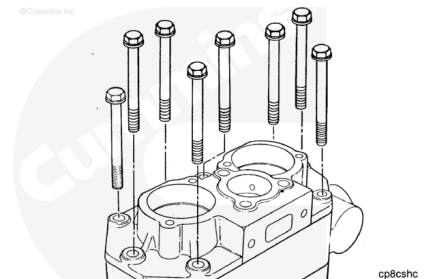


Install the cover.

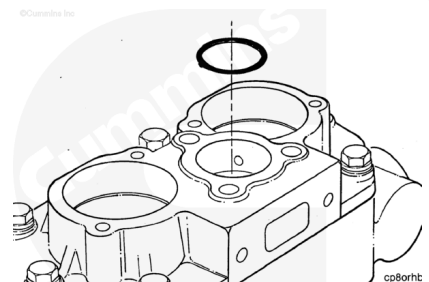


For the Holset ST676 air compressor, install the eight hexagon head capscrews and finger tighten.

For the Holset ST773 air compressor, install the eight captive washer capscrews.

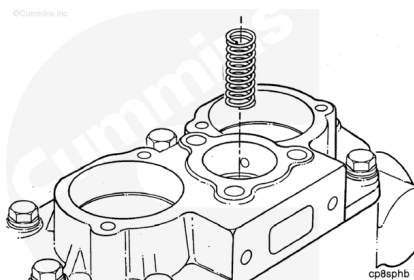


Use an o-ring pick, Part Number 3376399, or equivalent, to correctly install the new o-ring seal.



Note : Cummins Inc. recommends that new springs be installed during rebuild.

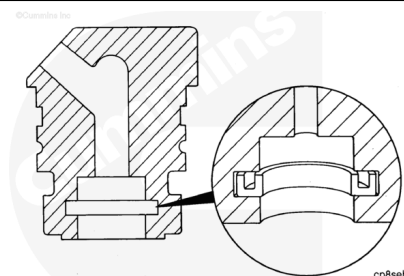
Install the unloading spring.



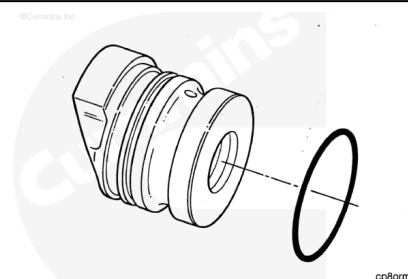
⚠ CAUTION ⚠

The rectangular ring seal must be installed with the grooved side up; failure to do so will result in air system damage and brake failure.

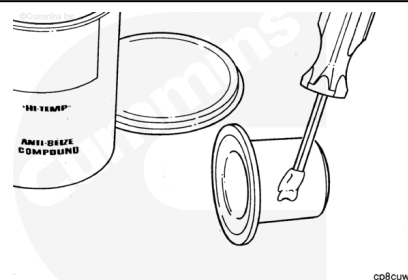
Use an o-ring pick, Part No. 3376399, to correctly install the new rectangular ring seal, if the unloader body has a cavity for this o-ring.



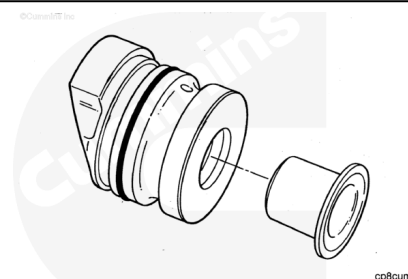
Install a new o-ring seal.



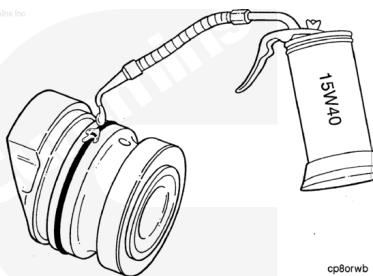
Use antiseize compound to lubricate the outside diameter of the unloader valve cap.



Install the cap into the unloading valve body.



Use clean 15W-40 lubricating engine oil to lubricate the seal.

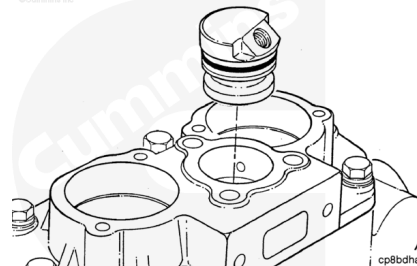


cp8orwb

⚠ WARNING ⚠

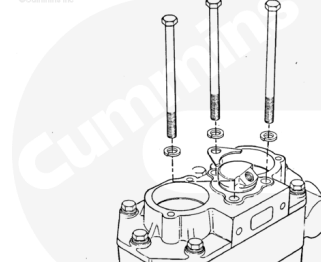
The unloader valve body is installed under spring tension. Carefully install to avoid personal injury. Wear face and eye protection.

Install the body into the cover.



cp8bdha

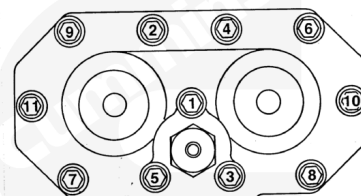
Install the retaining clamp and the three lock washers and the three hexagon head capscrews.



cp8cshd

Tighten the capscrews in the sequence shown.

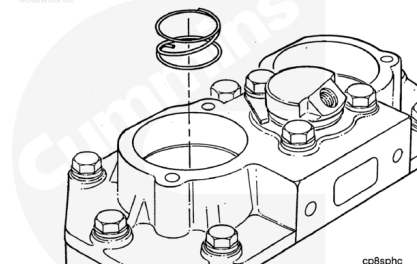
Torque Value: 41 n•m [30 ft-lb]



cp8csog

Note : Cummins Inc. recommends that new springs be installed during rebuild.

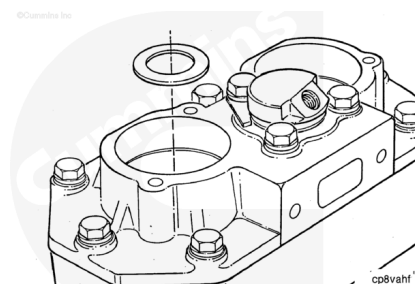
Install the intake valve spring with the tang end down.



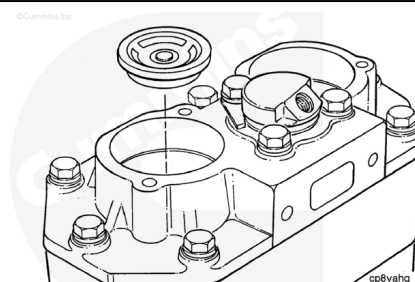
cp8sphc

Note : Cummins Inc. recommends that new intake valves be installed during rebuild.

Install the intake valve.

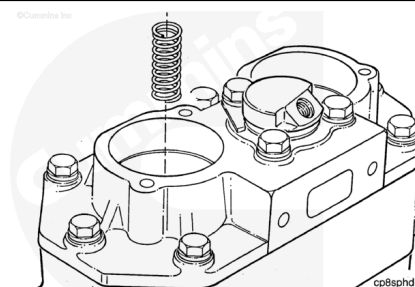


Install the intake valve seat with the flange end **up**.

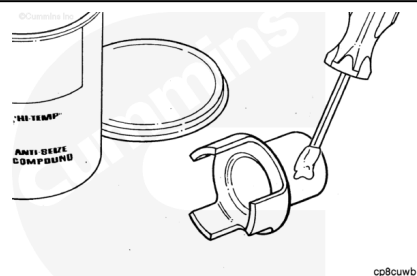


Note : Cummins Inc. recommends that new springs be installed during rebuild.

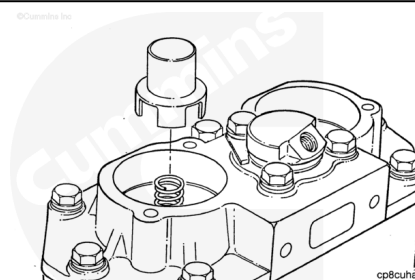
Install the unloader cap spring.



Use antiseize compound to lubricate the outside diameter of the unloader valve cap.

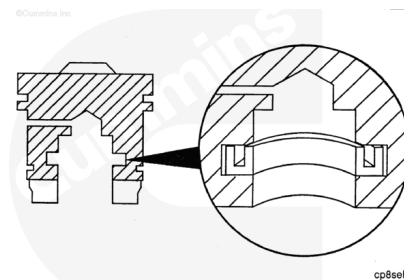


Install the cap into the cover and make sure the three tangs are in the three slots of the intake valve seat.



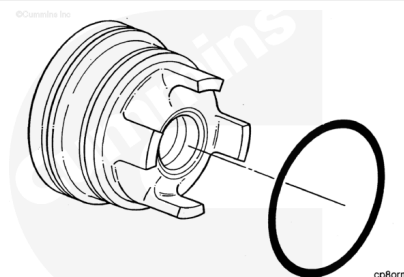
⚠ CAUTION ⚠

The rectangular ring seal must be installed with the grooved side up; failure to do so will result in air system damage and brake failure.

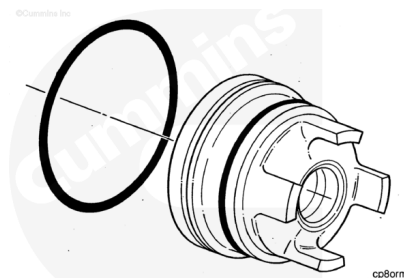


Use an o-ring pick, Part Number 3376399, or equivalent, to correctly install the new rectangular ring seal.

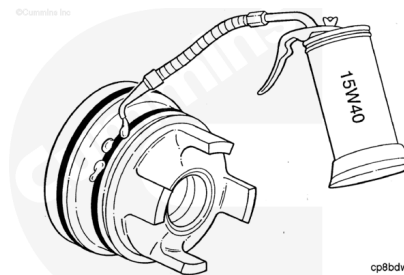
Install a new bottom o-ring seal.



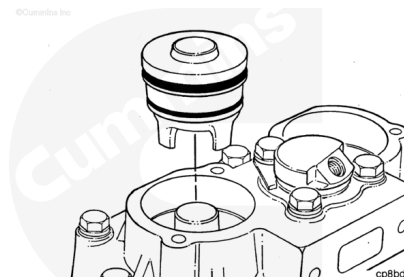
Install a new top o-ring seal.



Use clean 15W-40 lubricating engine oil to lubricate the seals.



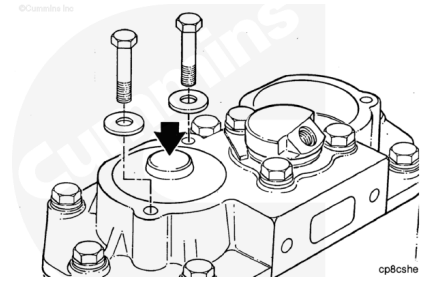
Install the unloading body into the cover.



⚠ WARNING ⚠

The unloader body is installed with spring tension. It must be installed as described to prevent personal injury.

Hold down on the unloading body firmly and install the capscrews and plain washers so as to prevent the unloader body from being thrown free and causing personal injury.



Tighten the capscrews.

Torque Value: 14 n•m [10 ft-lb]

Repeat the last thirteen steps to install the other unloading body assembly.

